

Multimodal AI Assistant for Medical Image and Clinical Text Analysis with Paired Image–Text Fusion

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Abstract—Medical information appears in images, scanned records, and typed clinical text, but these sources do not always belong to one patient cohort. This thesis therefore develops and evaluates three separate systems rather than presenting them as one fused pipeline. Model-1 is a standalone medical-image system for four-class Brain MRI classification and fourteen-label chest-radiograph classification. Leakage-aware grouped and patient-wise cross-validation produced strong internal performance for both tasks.

Model-2 is a standalone clinical-document and text system. Its scanned-document branch performs optical character recognition, rule-based field extraction, and structured export, while its doctor-typed-text branch performs medical-specialty classification and weak-label entity extraction. The evaluated pipeline completed OCR and structured export for all reported records and produced an independently held-out specialty result. These findings are interpreted only against their stated reference labels and are not treated as expert-verified semantic accuracy.

Model-3 is a separately trained architecture for paired Indiana University chest-radiograph studies. Within this architecture, a three-seed DenseNet-121 image ensemble and an Indication-only word- and character-level TF-IDF logistic-regression branch produce ten-label probabilities. Development-selected late fusion combines these probabilities, and a training-derived template module generates Findings text. On the locked 352-study test set, fusion produced the highest classification point estimates, and the generated Findings showed measurable lexical overlap with the hidden reference Findings. These results support bounded retrospective technical validation of three distinct systems; they do not establish clinical validity, external-hospital generalization, or deployment readiness.

Index Terms—Medical Image Analysis, Clinical Document Processing, Clinical Text Classification, Paired Image–Text Analysis, Late Fusion, Radiology Report Generation

I. INTRODUCTION

A. Background

Clinical information is recorded in several forms, including medical images, scanned documents, and typed narrative text. Each form requires a different analytical approach. Convolutional neural networks can identify visual patterns in radiolog-

ical images, whereas optical character recognition (OCR), text classification, and information-extraction methods are needed for prescriptions, laboratory reports, and clinical notes [1]–[3]. their outputs must be evaluated against task-appropriate reference data rather than combined into one unsupported performance claim.

Multimodal learning concerns the representation and use of more than one information source [4]. In medicine, meaningful multimodal evaluation requires the inputs to belong to the same patient or study and to share a defined target. A software interface that can execute several models does not by itself demonstrate valid multimodal reasoning. This distinction is central to the present work: Model-1 and Model-2 are standalone systems, whereas Model-3 is a separately trained paired-study architecture with its own image and text branches.

B. Rationale and Motivation

Medical images can reveal anatomical and pathological patterns, while scanned records and typed clinical text contain contextual and administrative information that is not available from pixels alone. Developing separate systems for these sources allows each task to use an appropriate dataset, model, target, and validation protocol. It also prevents the performance of one task from being used to imply the validity of another.

The initial implementation demonstrated image analysis and clinical-text processing, but an earlier multimodal claim lacked a genuinely paired validation cohort. the revised research therefore preserves Model-1 and Model-2 as independent systems and introduces Model-3 as a separate architecture. Model-3 is trained and evaluated only on same-study radiographs, Indication text, structured finding labels, and reference Findings. This design directly addresses the need for traceable validation while retaining the practical value of the individual image and text systems.

C. Problem Statement

Medical machine-learning results can be overstated when duplicate images or patients cross data partitions, when processing completion is mistaken for semantic correctness, or when unrelated image and text records are presented as a validated multimodal system. the research problem is to develop three technically useful but clearly separated systems and to match each claim to its actual cohort, reference target, split unit, and metric. Model-1 must be evaluated with leakage-aware image partitions; Model-2 must distinguish document-processing coverage from expert semantic accuracy; and Model-3 must use genuinely paired image and text inputs from the same radiographic study.

D. Objectives

The objectives of this thesis are:

- 1) To develop Model-1 as a standalone medical-image system for four-class Brain MRI and fourteen-label Chest X-ray classification;
- 2) To evaluate Model-1 through grouped Brain MRI and patient-wise Chest X-ray cross-validation, with model and threshold selection restricted to development data;
- 3) To develop Model-2 as a standalone system for scanned-document processing, medical-specialty classification, and weak-label entity extraction from doctor-typed text;
- 4) To report Model-2 results according to the quality of the available references, without treating completion or automatically generated labels as expert clinical validation;
- 5) To develop Model-3 as a separate paired IU architecture containing its own radiograph and Indication-text submodels, development-selected late fusion, and a training-derived Findings-generation module;
- 6) To evaluate Model-3 on a locked same-study test set using classification, uncertainty, and report-overlap measures; and
- 7) To document reproducibility controls, limitations, and the boundary between retrospective technical evaluation and clinical validation.

E. Methodology in Brief

Model-1A uses DenseNet-121 for Brain MRI classification and keeps exact and conservatively detected near-duplicate groups within the same cross-validation fold. Model-1B uses DenseNet-121 for multi-label Chest X-ray classification, with patient-wise outer folds and thresholds selected only from inner-validation patients. Model-2A applies page preprocessing, OCR, text cleaning, rule-based field extraction, and structured JSON export to scanned records. Model-2B applies text preprocessing and supervised specialty classification to doctor-typed text; its entity component is evaluated separately against weak labels.

Model-3 uses 2,287 eligible IU studies. A fixed development cohort of 1,935 studies is used for all training and selection, while 352 studies form a locked test set. Within Model-3,

a three-seed DenseNet-121 ensemble estimates image probabilities and a word- and character-level tF-IDF logistic-regression submodel estimates probabilities from Indication text. A development-selected late-fusion rule combines them. thresholded predictions are converted to Findings sentences through templates learned from the training data, and generated Findings are compared with the hidden same-study reference Findings.

F. Scope and Challenges

Model-1 is limited to the selected four-class Brain MRI and fourteen-label ChestXray14 tasks. Model-2 covers scanned prescriptions and laboratory reports, specialty classification for 29 retained MtSamples categories, and weak-label entity extraction. Model-3 is limited to ten findings in the eligible IU cohort. the datasets used by the three systems do not form one patient cohort, and Model-1 or Model-2 outputs are not inputs to Model-3.

the principal challenges are class imbalance, incomplete or automatically mined labels, unavailable patient identifiers in some sources, duplicate-image risk, variation in scanned-document quality, and limited expert annotation. the study is retrospective and internal. It does not establish prospective safety, independent-hospital generalization, patient benefit, or readiness for clinical deployment.

II. LITERATURE REVIEW

A. Preliminaries

This thesis contains three separate systems. Model-1 performs medical-image classification, Model-2 processes scanned documents and doctor-typed text, and Model-3 performs paired image-text analysis and Findings generation on Indiana University (IU) chest-radiograph studies. The first two systems are not input branches of the third. This separation is necessary because their datasets, prediction targets, and validation protocols are different.

For a single-label task, accuracy measures the proportion of correct predictions, while macro F1 gives equal importance to each class. For a multi-label task, each label is predicted independently. AUROC measures ranking across decision thresholds, whereas AUPRC is more sensitive to positive-class prevalence. Threshold-dependent F1 reflects the chosen operating point. These measures are not interchangeable, and values from different cohorts or split protocols do not form a direct leaderboard.

Data separation is especially important in medical imaging. If exact duplicates, visually similar images, or images from the same patient appear in both training and evaluation partitions, performance may be optimistic. The present work therefore uses similarity-grouped folds for Brain MRI and patient-wise folds for ChestXray14. Model-3 uses a fixed study-wise development/test partition because a stable patient identifier is unavailable in the accessed IU representation.

Text-generation quality is evaluated differently from diagnostic classification. ROUGE-1, ROUGE-2, and ROUGE-L measure unigram, bigram, and longest-common-subsequence

overlap between generated and reference text [5]. They provide a reproducible measure of lexical similarity but do not determine whether a report is clinically correct, complete, or safe.

B. Review of Existing Research

Medical-image classification. Dense convolutional networks are established baselines for medical-image analysis. In four-class Brain MRI classification, Rasheed et al. reported five-fold accuracy of 0.973 with DenseNet-121 on 7,023 images [6]. Disci et al. reported accuracy of 0.9285 for DenseNet-121, 0.9062 for ResNet-50, and 0.9527 for Xception under their supplied-folder protocol [7]. Mao et al. reported 0.499 accuracy for an original DenseNet-121 and 0.962 for an enhanced Dilated SE-DenseNet under a different ten-fold and test-time-cropping design [8]. These studies demonstrate strong image-classification performance, but differences in data versions, duplicate handling, pretraining, augmentation, and test construction limit direct numerical comparison.

For chest radiographs, Wang et al. introduced ChestXray8 and reported label-wise AUROC under a random image-level split [9]. Later work emphasized patient-level separation. Baltruschat et al. retained all images of a patient in the same partition and reported a mean pathology AUROC of 0.821 ± 0.012 for an image-only ResNet-50 configuration [10]. Kufel et al. used a patient-wise split and reported mean AUROC of 0.843 with an ImageNet-initialized EfficientNet-B1 [11]. The literature therefore supports both architecture comparison and strict reporting of the split unit. It also shows why AUROC should be accompanied by prevalence-aware and threshold-dependent measures in imbalanced multi-label tasks.

Clinical-document and text processing. OCR converts scanned pages into machine-readable text, but handwriting, page rotation, low contrast, stamps, and layout variation can cause errors. Ponnuru et al. combined preprocessing, Tesseract OCR, and regular expressions for medication extraction from prescriptions [3]. Datta et al. proposed a pipeline involving region localization, OCR, spelling correction, and structured storage for multilingual and handwritten prescriptions [12]. These studies support transparent staged processing, but an OCR call that completes successfully is not necessarily a correct transcription. Expert-corrected text and field annotations are needed for character error rate, word error rate, and semantic field accuracy.

Clinical-text classification and entity extraction also depend on the quality of their labels. BioBERT demonstrated the benefit of domain-specific pretraining for biomedical NLP [13]. The n2c2 medication-event task showed that clinical interpretation requires more than entity boundaries because negation, temporality, certainty, and action affect meaning [14]. Pagad et al. used clinical-text classification methods on MTSamples-derived data, but their class selection, preprocessing, and split differ from those used here [15]. These findings motivate a separate held-out evaluation for Model-2 specialty classification and a cautious interpretation of entity results derived from rules and dictionaries.

Paired multimodal analysis. Multimodal learning addresses representation, alignment, fusion, and co-learning across information sources [4]. A review by Huang et al. found that combining medical images with electronic health-record information can improve task-specific performance, while also identifying wide variation in datasets, fusion methods, and evaluation designs [1]. Zhi et al. combined CT and electronic-health-record features for pulmonary-embolism prediction and compared the multimodal model with its individual inputs [16]. Aksoy et al. added symptoms, vital signs, and notes to chest-radiograph report generation; automatic measures improved in some settings, although expert review still identified omissions [17]. The common methodological requirement is genuine alignment: the modalities must describe the same case and share a meaningful target.

The IU collection links de-identified chest-radiograph images with narrative reports and is widely used for report-generation research [18]. R2Gen introduced a memory-driven transformer and reported an IU ROUGE-L score of 0.371 [19]. The cross-modal memory network reported ROUGE-L of 0.375 on IU [20], while SERPENT-VLM later reported 0.452 under its own experimental configuration [21]. These papers establish ROUGE as a conventional report-generation measure, but their values are not directly comparable with a template-based system unless cohort construction, preprocessing, reference sections, tokenization, and metric implementation are matched.

Position of the present research. Model-1 follows established image architectures but strengthens internal validity through similarity-grouped and patient-wise evaluation. Model-2 uses a transparent document pipeline and a held-out specialty classifier, while explicitly separating processing coverage and weak-label agreement from expert semantic correctness. Model-3 is not a fusion of the standalone Model-1 and Model-2 systems. It is a new IU-specific architecture with independently trained image and Indication-text submodels, development-selected probability fusion, and training-derived Findings templates. Its classification and generation outputs are evaluated only on a locked same-study cohort.

C. Summary

Existing research supports three design decisions. First, medical-image results must be interpreted with the split unit and leakage controls used to obtain them. Second, document and text claims must reflect the quality of their reference annotations. Third, image-text fusion and report generation require genuinely paired studies and separate unimodal controls. The methodology in this thesis follows these principles while retaining Models 1, 2, and 3 as distinct systems with independent outputs.

III. REQUIREMENTS, IMPACTS AND CONSTRAINTS

A. System Specification

The research is specified as three independently bounded systems. Model-1 and Model-2 each produce standalone outputs; neither supplies an input to Model-3.

Table I
ARCHITECTURES, INPUTS, AND INDEPENDENT OUTPUTS OF THE THREE SYSTEMS.

System	Architecture and input	Individual output
Model-1	DenseNet-121 image classifiers; Brain MRI or Chest X-ray image.	Four-class Brain MRI probabilities or fourteen-label Chest X-ray probabilities and predictions.
Model-2	OCR and rule-based extraction for scans; text preprocessing and classification for doctor-typed text.	OCR text and structured fields, or specialty probabilities and weak-label entity fields.
Model-3	New IU-specific architecture: a three-seed DenseNet-121 radiograph ensemble and an Indication-only word/character TF-IDF logistic-regression submodel, followed by 0.75/0.25 late fusion and training-derived Findings templates.	Ten finding probabilities, thresholded findings, and generated Findings for one IU study.

The numbering identifies research components; it does not imply that all three were trained on one patient cohort. Each component has its own inputs, targets, and permitted interpretation.

B. Functional Requirements

- 1) Accept Brain MRI images and produce one of four class probabilities;
- 2) Accept Chest X-ray images and produce fourteen finding probabilities and thresholded labels;
- 3) Accept scanned prescriptions or laboratory reports and produce OCR text plus structured fields;
- 4) Accept doctor-typed text and produce specialty probabilities and weak-label entity fields;
- 5) Accept same-study IU radiographs and Indication text within Model-3 and generate ten-label probabilities and Findings text;
- 6) Preserve model probabilities, selected thresholds, fold identifiers, and source identifiers needed for evaluation; and
- 7) Export structured machine-readable results for reproducibility checks.

C. Non-functional Requirements

1) *Reproducibility*: Dataset manifests, split assignments, random seeds, checkpoints, and out-of-fold predictions must be retained separately. Model or threshold selection must use training or development data only. Final-test labels must not influence architecture, fusion-weight, or threshold decisions.

2) *Modularity*: Image, document, typed-text, and paired-fusion code paths must remain separable. A failure in one branch must not be represented as evidence about another. This structure also prevents a component completion measure from being reported as end-to-end accuracy.

3) *Privacy and Security*: The prototype must use de-identified or public research data in the reported experiments. Identifiable examples must not appear in the report. Any future use with clinical records would require access control, encrypted storage, audit logging, retention policies, and institutional approval.

4) *Usability and Traceability*: Outputs should identify their source branch, version, probability or extracted field, and applicable warning. The interface is intended for technical inspection, not autonomous clinical use.

D. Evidence and Acceptance Criteria

Table II defines what constitutes evidence for each component.

Table II
COMPONENT ACCEPTANCE CRITERIA AND INTERPRETATION BOUNDARIES.

Component	Acceptance evidence	Excluded interpretation
Model-1A	Complete out-of-fold predictions under grouped five-fold validation; accuracy, macro F1, AUROC, calibration, and confusion analysis.	Patient-level or external-institution validity when patient identifiers and external data are unavailable.
Model-1B	Complete patient-wise out-of-fold predictions; AUROC and thresholded F1 with inner-only threshold selection.	Performance from random image splits or thresholds chosen on the outer test fold.
Model-2A	Document counts, OCR execution, field-production coverage, and structured-output completion.	Semantic extraction accuracy without expert-corrected text or field annotations.
Model-2B	Held-out specialty classification and weak-label entity metrics.	Expert clinical entity recognition or general typed-text understanding.
Model-3	Locked same-study image-text evaluation with unimodal baselines, class-imbalance-aware metrics, and paired uncertainty.	General multimodal validity across MRI, documents, typed records, or different hospitals.

E. Societal and Ethical Impact

A modular research workflow may reduce the technical effort needed to compare image, document, and text-processing methods. The present work does not evaluate time savings, user benefit, or clinical outcomes, so those benefits remain hypotheses.

The principal risks are privacy loss, dataset bias, false confidence, and misuse of model output. Public datasets may not represent the patient populations, equipment, documentation styles, or disease prevalence of another institution. Weak labels may encode the errors of the rules that created them. Imbalanced datasets can also make accuracy appear high even when positive cases are missed.

Professional responsibility requires that these limits remain visible. The models are not medical devices, their outputs are

not diagnoses, and the prototype cannot replace a clinician. Future human-subject research would require ethics review, informed governance, de-identification, qualified reference annotation, and predefined safety outcomes.

F. Environmental and Economic Constraints

The main environmental cost is electricity consumed during GPU training and storage of repeated checkpoints. The five-fold image experiments are the most computationally intensive part of the study. Reusing frozen splits, retaining valid checkpoints, and avoiding unnecessary repeat training reduce this cost.

The research uses open-source software and local hardware, so direct software expenditure is limited. Its real costs are GPU time, electricity, storage, development effort, and dataset preparation. Deployment economics are outside scope because a production system would also require security engineering, integration, regulatory work, maintenance, and expert validation.

G. Standards and Regulatory Position

The study does not claim compliance with medical-device, hospital-integration, or information-security certification. Its design nevertheless follows relevant engineering principles: separation of development and test data, versioned artifacts, provenance, privacy-aware reporting, and explicit human oversight. Formal deployment would require institution-specific legal and regulatory assessment in addition to technical validation.

H. Research Work Management

The work was organized into the phases shown in Table III.

Table III
RESEARCH PHASES AND PRINCIPAL OUTPUTS.

Phase	Principal output
Problem definition	Clinical problem, scope, and literature review.
Data preparation	Dataset manifests, integrity audits, and preprocessing pipelines.
Component development	Image classifiers, document processor, and typed-text modules.
Evaluation correction	Grouped Brain MRI folds and patient-wise Chest X-ray folds.
Paired validation	IU development/locked-test cohort and image-text fusion experiment.
Reporting	Result verification, claim audit, bibliography review, and final document preparation.

Responsibilities were divided among team members by component, while dataset definitions, final metrics, and report terminology were checked across the complete research work. Outputs from later experiments were stored separately to avoid overwriting selected checkpoints or prior evidence.

I. Risk Management

Table IV
PRINCIPAL RESEARCH RISKS AND CONTROLS.

Risk	Consequence	Control
Duplicate or patient leakage	Inflated image performance.	Group Brain MRI similarity candidates; split Chest X-rays by patient; verify zero cross-fold overlap.
Outer-test tuning	Optimistic generalization estimate.	Select checkpoints, thresholds, and fusion weight from inner or development predictions only.
OCR and weak-label error	Incorrect text or fields presented as truth.	Report completion separately from correctness and disclose the annotation source.
Class imbalance	Misleading accuracy and poor positive recall.	Report AUROC, AUPRC, F1, balanced accuracy, support, and an all-negative control where relevant.
Unpaired modalities	Unsupported multimodal claim.	Restrict Model-3 evidence to same-study IU image-text pairs.
Over-trust	Prototype output used as clinical advice.	State non-clinical scope in the abstract, methods, results, and conclusion.
Privacy exposure	Disclosure of sensitive information.	Use public/de-identified data and omit identifiable examples.

IV. METHODOLOGY AND DESIGN PROCESS

A. Design Overview

The research contains three separate systems with independent outputs. Model-1 analyzes medical images, Model-2 processes scanned documents and doctor-typed text, and Model-3 is a new paired-study architecture developed for the Indiana University (IU) chest-radiograph cohort. Model-1 and Model-2 are not fused, linked, or reused as input branches of Model-3. Within Model-3, the image and Indication-text submodels are trained specifically on the IU development cohort before their probabilities are combined.

This separation follows the available evidence. Model-1 has image-specific reference labels and leakage-aware folds; Model-2 has document-processing and text-specific references; Model-3 has genuinely paired radiographs, Indication text, structured finding labels, and reference Findings from the same study. Consequently, every reported output can be traced to the cohort and target used to evaluate it.

B. Model-1: Standalone Medical-Image System

Model-1 contains two image-classification tasks under one standalone image-analysis system. Model-1A predicts four Brain MRI classes: glioma, meningioma, no tumor, and pituitary tumor. Model-1B predicts fourteen Chest X-ray findings. Both tasks use DenseNet-121, but they have different output activations and validation units. Model-1A applies softmax to

produce one four-class distribution, while Model-1B applies sigmoid activation to produce fourteen independent probabilities.

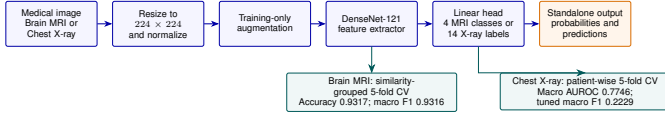


Figure 1. Model-1 standalone medical-image architecture and its task-specific validation protocols.

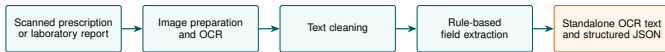
For Brain MRI, images are resized to 224×224 , normalized, and augmented only during training. Exact hashes and conservative perceptual-similarity links define groups that remain within one fold. DenseNet-121 and ResNet-50 are also trained under matched grouped folds to provide an internal architecture comparison. For Chest X-ray, patient identifiers define the outer and inner partitions. Decision thresholds are selected from inner-validation patients and then applied unchanged to the corresponding outer fold.

The output of Model-1 is self-contained: an input image produces probabilities and the associated predicted class or labels. These outputs are not passed to Model-2 or Model-3.

C. Model-2: Standalone Document and Text System

Model-2 also has two distinct paths. Model-2A accepts a scanned prescription or laboratory report. Page preparation improves orientation and readability, OCR produces machine-readable text, text cleaning reduces formatting noise, and rule-based extraction maps supported patterns to structured fields. Model-2B accepts doctor-typed clinical text directly, without OCR. It performs specialty classification and runs a separate weak-label entity component.

Model-2A: scanned-document processing



Model-2B: doctor-typed-text analysis

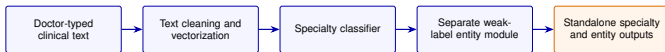


Figure 2. Model-2 standalone scanned-document and doctor-typed-text architecture.

The two paths have separate interpretations. Model-2A processing completion shows whether OCR, field production, and structured export executed; it does not establish that every recognized character or extracted value is correct. Model-2B specialty classification is evaluated on a held-out split, whereas entity extraction is evaluated only against labels generated by rules and dictionaries. Model-2 therefore produces its own document or text output and does not provide the text branch used by Model-3.

D. Model-3: Paired IU Architecture

Model-3 is built and trained independently for the paired IU experiment. One eligible study supplies two predictor sources:

all available radiographic projections and the same report’s Indication field. The reference Findings and Impression sections are excluded from every model input. The image submodel is a three-seed ImageNet-pretrained DenseNet-121 ensemble. Its view-level predictions are aggregated into ten study-level probabilities. The text submodel uses word- and character-level TF-IDF features with ten one-vs-rest logistic-regression classifiers.

The two probability vectors refer to the same ten findings and are combined through development-selected late fusion:

$$p_k^{\text{fusion}} = 0.75p_k^{\text{image}} + 0.25p_k^{\text{text}}, \quad (1)$$

where k denotes one of the ten fixed IU labels. Thresholds selected from development out-of-fold predictions convert the fused probabilities into predicted findings. A deterministic template module, whose label-to-sentence mappings are constructed from the training data, converts the predicted label set into generated Findings text. This module does not retrieve or copy a test report.



Figure 3. Model-3 architecture. Its IU image and Indication-text submodels are trained specifically for this system; the reference Findings are used only after prediction for locked-test evaluation.

The generated Findings are compared with the actual Findings from the same locked test study using ROUGE-1, ROUGE-2, and ROUGE-L. Classification metrics are reported separately because they evaluate ten-label diagnostic prediction, whereas ROUGE measures lexical overlap between two texts. Neither metric alone establishes clinical correctness.

E. Input, Output, and Validation Boundaries

Table V
INPUTS, OUTPUTS, AND EVALUATION BOUNDARIES OF THE THREE SYSTEMS.

System	Input	Individual output	Primary evaluation boundary
Model-1	Brain MRI or Chest X-ray image.	Class/label probabilities and predictions.	Similarity-grouped Brain MRI folds or patient-wise Chest X-ray folds.
Model-2	Scanned medical document or doctor-typed text.	OCR/structured fields or specialty/entity output.	Processing coverage, held-out specialty labels, or disclosed weak labels.
Model-3	Same-study IU radiographs and Indication text.	Ten-label probabilities, predicted findings, and generated Findings.	Development-only selection followed by one locked 352-study evaluation.

Model-1 and Model-2 provide internal technical evidence only for their own tasks. Model-3 supports a retrospective same-study comparison of image-only, Indication-only, and fused prediction, followed by template-based Findings generation. None of the three systems has undergone prospective clinical validation or independent-hospital testing.

Chapter Summary

The final design keeps all three systems independent. Model-1 produces medical-image classifications, Model-2 produces document and typed-text outputs, and Model-3 contains its own IU-specific image and text submodels, late fusion, and Findings-generation stage. The next chapter describes the datasets and partition rules used to evaluate these architectures.

V. DATA COLLECTION AND ANALYSIS

A. Dataset Roles and Provenance

The research uses separate datasets because no single source contains all required modalities and reference targets. Table VI records the source and role of each dataset. Counts in later sections refer to the local manifests used in the reported experiments, not necessarily to every item available at the source.

Table VI
DATASET PROVENANCE AND EXPERIMENTAL ROLE.

Component	Source	Role
Brain MRI	Four-class Brain Tumor MRI collection [22]	Model-1A training, grouped cross-validation, and matched architecture comparison.
Chest X-ray	NIH ChestXray14 images and metadata [9], [23]	Model-1B patient-wise multi-label training and evaluation.
Scanned documents	Prescription and medical-report resources [24]–[26]	Model-2A OCR and rule-based field-processing evaluation.
Doctor-typed text	MTSamples medical-transcription collection [27]	Model-2B specialty classification and weak-label entity evaluation.
Paired image–text	Indiana University chest-radiograph/report collection [18], [28]	Model-3 same-study image, Indication-text, and fusion comparison.

B. Model-1A Brain MRI Dataset

The local Brain MRI manifest contains 7,200 readable images: 5,600 in the supplied Training folder and 1,600 in Testing. The four classes are glioma, meningioma, no tumor, and pituitary tumor. No unreadable image or conflicting label within a detected duplicate group was found.

1) *Duplicate and Similarity Audit*: Cryptographic hashes identified 153 exact-duplicate groups; none crossed the supplied folder boundary. A conservative 64-bit perceptual-hash search at Hamming distance at most four found 5,689 non-identical candidate pairs, including 2,045 pairs that crossed the supplied boundary. Combining exact and candidate links produced 1,103 groups. Of these, 516 crossed the original boundary, and 3,554 images belonged to a group.

Perceptual similarity is a screening method, not proof that two files are the same acquisition. The candidates were therefore not deleted. Instead, the complete connected group became the partition unit. This choice may be conservative, but it prevents identified candidates from appearing in both development and evaluation data.

2) *Grouped Partition Construction*: Five outer folds were created with seed 42 using stratification by class and grouping by the combined duplicate/similarity identifier. A grouped inner-validation subset was drawn only from the outer-training groups. Each image receives one out-of-fold prediction, and no detected group crosses an outer or inner boundary.

C. Model-1B ChestXray14 Dataset

The metadata audit matched all 112,120 rows to image files and identified 30,805 patients. Fourteen pathology labels are retained: atelectasis, cardiomegaly, effusion, infiltration, mass, nodule, pneumonia, pneumothorax, consolidation, edema, emphysema, fibrosis, pleural thickening, and hernia. The *No Finding* field is not treated as a fifteenth pathology.

1) *Patient-Overlap Audit*: The earlier seed-42 random image split assigned 89,696 images to training and 22,424 to evaluation. It placed 8,022 patients in both sets; 69.00% of

evaluation patients also appeared in training. That split was discarded as generalization evidence.

For the corrected evaluation, patient-level multi-label vectors were assigned to five outer folds. A second patient-wise split within each outer-training set produced the inner-validation partition. Both AP and PA images were retained. The final manifests contain no duplicate image identifier, missing image, empty label row, unknown label token, or patient overlap between training, validation, and outer evaluation.

D. Model-2A Scanned Documents

The Model-2A evaluation manifest contains 241 processed prescription and laboratory-report records. Pages vary in layout, contrast, typeface, handwriting, stamps, and table structure. OCR converts each image to text; rule-based patterns then search for supported fields. The available data do not include a complete expert-corrected transcription and field-level reference set for these exact 241 outputs.

For that reason, the dataset supports three bounded measures: whether OCR returned output, whether at least one expected entity field was produced, and whether a structured record was saved. It does not support word error rate or semantic field accuracy.

E. Model-2B Doctor-Typed-Text Dataset

The local MTSamples-derived table contains 4,999 raw rows, of which 4,966 remained after removing unusable text. The specialty classifier retained 4,841 typed records across 29 specialties. The split contains 2,904 training, 968 validation, and 969 test records. An additional inference manifest contains 250 processed records for interface and output checks.

Entity labels were created from rules and dictionaries rather than clinician annotation. This property affects the task definition: reported entity F1 measures weak-label agreement. The high proportion of non-entity tokens also makes token accuracy unsuitable as the primary result.

F. Model-3 Multimodal Paired Cohort

The peer-reviewed Indiana University collection describes approximately 3,996 reports and 8,121 linked images [18]. The accessed derivative exposed 3,851 report rows and 7,466 projection images before local integrity filtering [28]. These figures describe different representations and are reported separately.

Task-specific filtering produced 2,287 usable studies with linked radiographs, Indication text, and eligible structured labels. A fixed seed-2026 study-wise split assigned 1,935 studies to development and 352 to the locked final test. Development data alone supplied five out-of-fold partitions for configuration and threshold selection.

Table VII
IU PAIRED COHORT AND EXPERIMENTAL USE.

Representation	Count	Use
Published source description	3,996 reports / 8,121 images	Collection context.
Accessed derivative	3,851 rows / 7,466 projections	Local pre-filter representation.
Usable paired cohort	2,287 studies	Fixed experimental cohort.
Development cohort	1,935 studies	All model, fusion-weight, and threshold selection.
Locked final test	352 studies	One evaluation after freezing the configuration.

1) *Labels and Predictor Text*: Ten labels were fixed before final evaluation: Cardiomegaly, Pulmonary Atelectasis, Calcified Granuloma, Cicatrix, Pleural Effusion, Atherosclerosis, Airspace Disease, Scoliosis, Granulomatous Disease, and Nodule. Labels come from the structured Problems field. Indication is the only predictor text. Findings is the hidden primary reference for report-generation evaluation, and Impression is retained only as a secondary hidden reference; neither section is used as a predictor input.

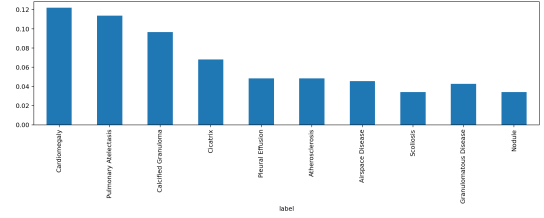


Figure 4. Prevalence of the ten fixed IU labels. Rare positive classes motivate AUPRC, F1, balanced accuracy, and per-label reporting.

2) *Label-Quality Audit*: A frozen synonym dictionary and negation-aware matching procedure compared development Findings with the structured labels for quality auditing. Conservative structured-positive negation rates ranged from 0 to 0.0309. Structured-negative/Findings-positive counts ranged from 1 for Scoliosis to 299 for Pleural Effusion. The structured labels were not rewritten after this audit. The discrepancies indicate incomplete or differently scoped labels and bound the interpretation of rare-class results.

3) *Pairing and Leakage Limits*: Every Model-3 image and Indication pair belongs to the same radiographic study. A stable patient identifier was unavailable in the accessed derivative, so separation is study-wise rather than patient-wise. Unrecognized repeat patients cannot be excluded. This residual risk is distinct from the ChestXray14 experiment, for which patient identifiers are available.

G. Preprocessing

1) *Images*: Brain MRI and ChestXray14 images are resized to 224×224 and normalized. Horizontal flips and small

rotations are applied to training images only. Evaluation images are not augmented. IU projections are also resized to 224×224 ; study-level probabilities are aggregated across available views using weights selected on development predictions.

2) *Scanned Documents*: Document preprocessing standardizes orientation and image quality where supported by the pipeline, then applies OCR. Text cleaning removes non-informative formatting symbols and combines fragmented lines. Rules preserve uncertain or missing values rather than inventing replacements.

3) *Doctor-Typed Text and IU Indications*: Doctor-typed records are filtered for usable text and mapped to retained specialties. IU Indications preserve negation and are represented with concatenated word- and character-level TF-IDF features. The text vectorizer and classifier are fitted on development data only. Label-to-sentence templates for Model-3 Findings generation are derived from training data and then frozen before locked-test evaluation.

Chapter Summary

Dataset provenance and split units determine what the models can establish. The two Model-1 datasets permit leakage-aware internal evaluation, Model-2 has bounded document and weak-label text evidence, and Model-3 has genuine study-level pairing but no reliable patient identifier. Chapter 6 specifies the implementation and metrics used within those limits.

VI. EXPERIMENTAL SETUP AND REPRODUCIBILITY

A. Development Environment

The experiments were implemented in Python on a local workstation with an NVIDIA RTX 5090 GPU. PyTorch, Torchvision, and `timm` support the image experiments. OCR and image preprocessing use local document-processing libraries and OpenCV, while structured outputs are stored as JSON and tabular result files. A dedicated GPU environment isolates the required dependencies.

Table VIII
IMPLEMENTATION ENVIRONMENT BY TASK.

Task	Implementation
Image training and inference	Python, PyTorch, Torchvision, <code>timm</code> , CUDA-enabled GPU.
Document processing	OCR, OpenCV preprocessing, rule-based field extraction, JSON export.
Doctor-typed-text analysis	Text preprocessing, specialty classifier, weak-label entity evaluation.
IU text model	Word- and character-level TF-IDF with one-vs-rest logistic regression.
Evaluation	Persisted fold manifests, probabilities, thresholds, metrics, plots, and checkpoints.

B. Model-1 Experimental Setup

1) *Common Image Handling*: Images are resized to 224×224 and normalized. Training-only augmentation consists of horizontal flips and rotations. DenseNet-121 is initialized randomly for the Brain MRI and ChestXray14 experiments to match the verified implementation. The best checkpoint

is selected from inner-validation performance, then evaluated once on the outer fold.

2) *Brain MRI Grouped Five-Fold Protocol*: Five outer folds are generated with seed 42. Stratified grouped splitting balances the four classes while keeping every exact/perceptual-similarity component within one fold. A grouped inner-validation subset is drawn from the remaining four folds.

Training uses cross-entropy loss, AdamW with learning rate 10^{-4} , batch size 64, at most 20 epochs, and early-stopping patience 4. DenseNet-121 and ResNet-50 use identical fold assignments, preprocessing, augmentation, optimizer, epoch budget, and stopping rule. This matched protocol permits a paired case-level comparison across all 7,200 out-of-fold predictions.

3) *Chest X-ray Patient-Wise Five-Fold Protocol*: Patient-level label vectors are assigned to five outer folds with iterative multi-label stratification and seed 42. Within each outer-training set, a second patient-wise split supplies inner validation. The evaluation covers all 112,120 images from 30,805 patients.

DenseNet-121 is trained with class-weighted binary cross-entropy with logits, AdamW with learning rate 5×10^{-5} , cosine learning-rate decay, batch size 64, at most 20 epochs, and early-stopping patience 4. Per-label thresholds are selected by inner-validation F1 over the grid 0.05 to 0.95 in steps of 0.05. The selected thresholds are then applied unchanged to the untouched outer fold. Results at threshold 0.5 are retained separately.

C. Model-2 Implementation

1) *Scanned-Document Pipeline*: Each page passes through image preparation, OCR, text cleaning, rule matching, and structured export. The record stores source metadata, extracted OCR text, supported fields, warnings, and execution status. A run counts as OCR-complete when a text result is stored, entity-complete when the expected structured field container contains extracted content, and JSON-complete when the record is serialized successfully.

These definitions are deliberately mechanical. The absence of expert-corrected transcriptions and field labels prevents calculation of character error rate, word error rate, or semantic field accuracy for the reported set.

2) *Doctor-Typed-Text Pipeline*: Usable MTSamples text is mapped to one of 29 retained medical specialties. The fixed split contains 2,904 training, 968 validation, and 969 test records. Accuracy, macro F1, and weighted F1 are calculated on the held-out test set.

The entity component is evaluated separately. Rules and dictionaries generate its reference labels, so the experiment reports token accuracy and entity F1 against weak labels. Entity F1 is prioritized because the large number of non-entity tokens can dominate token accuracy.

D. Model-3 Paired Experimental Protocol

1) *Image Branch*: The IU image branch is a three-seed ensemble (42, 123, and 2026) of ImageNet-pretrained DenseNet-121 networks. Each final seed follows a fixed 16-epoch staged

schedule: two classifier-head epochs, eight late-block epochs, and six full-network epochs. Binary cross-entropy is used for the ten labels.

Development experiments considered chest-X-ray-specific pretraining, asymmetric loss, 320×320 and 384×384 inputs, and learned gated multi-view aggregation. These alternatives did not improve the frozen development ranking. Per-label frontal/view aggregation learned from development predictions was retained. The development ensemble changed macro AUPRC from 0.2470 to 0.2504 and macro AUROC from 0.7691 to 0.7779; macro F1 changed from 0.3216 to 0.3211.

2) *Text Branch*: The text branch uses Indication only and preserves negation. Concatenated word- and character-level TF-IDF features feed ten one-vs-rest logistic-regression classifiers. Regularization $C = 0.25$ was selected on development out-of-fold predictions, where the text branch achieved macro AUPRC 0.1097.

3) *Calibration, Fusion, and Threshold Selection*: Platt calibration improved Brier-score and expected-calibration-error diagnostics but did not improve the ranking criteria, so uncalibrated probabilities were retained. Global fusion, per-label fusion, stacking, calibrated global fusion, and selective fusion were compared on development predictions. The selected global image weight was $\alpha = 0.75$, chosen first by macro AUPRC, then macro AUROC and macro F1 as tie-breakers.

Per-label decision thresholds were also selected from development out-of-fold predictions. Neither α nor the thresholds were adjusted after opening the locked test.

4) *Findings-Generation Procedure*: After thresholding, the predicted ten-label vector is converted into Findings text through a deterministic template module. Candidate label-to-sentence wording is derived from the Model-3 training records and frozen before evaluation. The generator receives only predicted labels; it does not receive, retrieve, or copy the locked study’s Findings or Impression. Generated Findings are then aligned with the actual same-study Findings solely for evaluation.

E. Evaluation Metrics

1) *Single-label Classification*: Brain MRI evaluation reports accuracy, macro F1, one-versus-rest macro AUROC, multiclass Brier score, expected calibration error, and a pooled out-of-fold confusion matrix. Macro F1 gives each class equal weight.

2) *Multi-label Classification*: Chest X-ray and IU evaluation report macro and micro AUROC, AUPRC where available, macro and micro F1, precision, recall, balanced accuracy, and per-label support. AUROC is threshold-independent; F1 and exact match depend on the chosen threshold. Label-wise accuracy is presented with an all-negative baseline for the sparse IU task.

3) *Document and Text Processing*: Model-2A reports processing coverage only. Model-2B specialty classification reports accuracy and class-balanced F1 measures. Entity evaluation reports entity F1 against weak labels and includes token

accuracy only to show why a negative-dominated metric can be misleading.

4) *Generated-Text Overlap*: ROUGE-1, ROUGE-2, and ROUGE-L compare generated Findings with the reference Findings [5]. The first two measure unigram and bigram overlap; ROUGE-L is based on the longest common subsequence. ROUGE is reported separately from classification metrics because lexical overlap does not test diagnostic discrimination or guarantee clinical correctness.

F. Statistical Analysis

Brain MRI fold means are summarized with sample standard deviations and 95% t -intervals. Chest X-ray fold means use sample standard deviations and fold-bootstrap intervals. The DenseNet-121 versus ResNet-50 comparison uses paired bootstrap resampling of matched out-of-fold cases.

For Model-3, paired case bootstrap resampling estimates confidence intervals for fusion-minus-image and fusion-minus-text differences. An interval crossing zero is interpreted as insufficient evidence of a consistent difference in the locked cohort, even when one point estimate is higher.

G. Reproducibility Controls

- 1) Fixed seeds are recorded for split construction and the IU image ensemble;
- 2) Outer-fold, inner-validation, and locked-test manifests are stored separately;
- 3) Selected thresholds and fusion weights are derived from development predictions only;
- 4) Each evaluation case retains a source or study identifier and its prediction vector;
- 5) Fold checkpoints and result files are kept in separate directories to avoid silent overwriting;
- 6) Aggregate metrics are recomputed from persisted predictions rather than entered manually; and
- 7) The template mapping and ROUGE evaluation configuration are frozen and stored with the Model-3 outputs.

Chapter Summary

The experimental setup uses nested separation for the two Model-1 tasks and separate development and locked-test cohorts for Model-3. Model-2 results remain tied to their actual reference quality. Model-3 classification and Findings-generation procedures are frozen before the locked evaluation and are reported as distinct evidence types in the next chapter.

VII. RESULTS AND EVALUATION

A. Evaluation Overview

Results are organized by system and evidence type. Image cross-validation, document processing, typed-text classification, paired prediction, and Findings generation are not reduced to one overall accuracy because their targets and reference standards differ.

Table IX
PRINCIPAL RESULTS AND INTERPRETATION BOUNDARIES.

Component	Result	Interpretation
Model-1A Brain MRI	Grouped five-fold accuracy 0.9317 ± 0.0168 ; macro F1 0.9316 ± 0.0169 ; 7,200 out-of-fold predictions.	Primary internal estimate with detected duplicate/similarity groups confined to folds.
Model-1B Chest X-ray	Patient-wise five-fold macro AUROC 0.7746 ± 0.0044 ; tuned macro F1 0.2229 ± 0.0031 ; 112,120 out-of-fold predictions.	Primary internal estimate with zero patient overlap and inner-only threshold selection.
Model-2A documents	241 records; OCR completion 1.0000; entity-field completion 0.8755; structured-output completion 1.0000.	Processing coverage only; no expert semantic-accuracy reference.
Model-2B specialty	Test accuracy 0.3209; macro F1 0.3468; weighted F1 0.2962.	Held-out specialty classification on the stated MTSamples-derived split.
Model-2B entities	Token accuracy 0.9937; entity F1 0.0148.	Agreement with rule/dictionary weak labels; not expert clinical entity recognition.
Model-3 paired fusion	Locked-test macro AUROC/AUPRC/F1: image 0.7910/0.3064/0.2915; text 0.6282/0.1257/0.1442; fusion 0.7954/0.3107/0.3071.	Same-study internal validation for ten IU labels; fusion point estimates are higher, but fusion-minus-image intervals cross zero.

B. Model-1A Brain MRI Results

DenseNet-121 completed all five grouped outer folds. Every one of the 7,200 images received one out-of-fold prediction, and no detected exact/perceptual-similarity group crossed an outer or inner partition. Mean accuracy was 0.9317 with standard deviation 0.0168 and a 95% t -interval of 0.9108–0.9526. Mean macro F1 was 0.9316, and mean one-versus-rest macro AUROC was 0.9905.

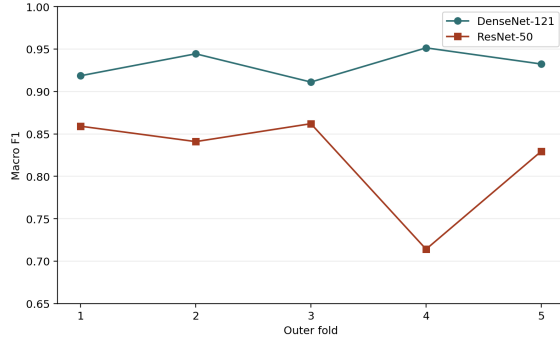


Figure 5. Model-1A DenseNet-121 performance across five grouped outer folds.

Table X
MODEL-1A GROUPED FIVE-FOLD DENSENET-121 RESULTS.

Fold	Images	Accuracy	Macro F1	Macro AUROC	Brier	ECE
1	1,440	0.9181	0.9185	0.9894	0.1193	0.0330
2	1,440	0.9444	0.9444	0.9908	0.0891	0.0233
3	1,440	0.9118	0.9111	0.9893	0.1316	0.0305
4	1,440	0.9514	0.9514	0.9920	0.0788	0.0185
5	1,440	0.9326	0.9324	0.9910	0.0963	0.0266
Mean $\pm$ SD	7,200 OOF	0.9317 ± 0.0168	0.9316 ± 0.0169	0.9905 ± 0.0011	0.1030 ± 0.0218	0.0262 ± 0.0055
95% t -interval	–	[0.9108, 0.9526]	[0.9105, 0.9526]	[0.9891, 0.9919]	[0.0759, 0.1301]	[0.0194, 0.0330]

Table XI
CONTROLLED BRAIN MRI ARCHITECTURE COMPARISON ON MATCHED GROUPED FOLDS.

Architecture	Parameters	Accuracy	Macro F1	Macro AUROC
DenseNet-121	6,957,956	0.9317 ± 0.0168	0.9316 ± 0.0169	0.9905 ± 0.0011
ResNet-50	23,516,228	0.8253 ± 0.0571	0.8211 ± 0.0613	0.9545 ± 0.0254
Paired difference	–	0.1064	0.1098	–
Paired bootstrap 95% CI	–	[0.0981, 0.1153]	[0.1013, 0.1182]	–

Table XII
PROTOCOL-AWARE CONTEXT FOR PUBLISHED FOUR-CLASS BRAIN MRI RESULTS.

Study	Protocol	Reported result	Reason not directly comparable
This thesis	Five grouped outer folds; detected similarity groups confined to folds; random initialization.	Accuracy 0.9317; macro F1 0.9316.	Internal reference; patient identifiers and external data unavailable.
Rasheed et al. [6]	Five-fold analysis plus separately described 80/20 split; ImageNet initialization.	Accuracy 0.973.	Duplicate control and relation between the two evaluations are not fully reported.
Disci et al. [7]	Supplied folders; transfer learning; testing augmentation reported.	DenseNet 0.9285; ResNet 0.9062; Xception 0.9527 accuracy.	Split, initialization, preprocessing, and testing policy differ.
Mao et al. [8]	Ten-fold validation; non-pretrained models; ten-crop inference.	DenseNet 0.499; enhanced model 0.962 accuracy.	Architecture and inference protocol differ; duplicate grouping unreported.

The pooled confusion matrix contains predictions from an outer fold for every image. The most frequent error was glioma classified as meningioma (126 cases). Pooled class F1 was 0.9247 for glioma, 0.8996 for meningioma, 0.9459 for no tumor, and 0.9563 for pituitary tumor.

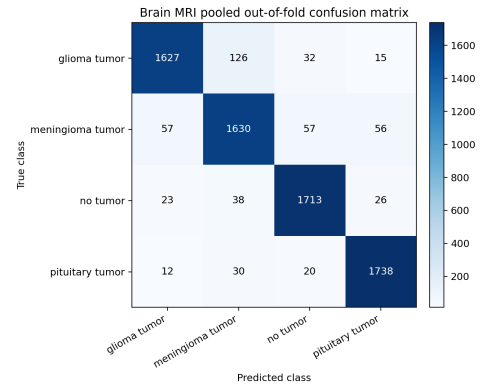


Figure 6. Pooled out-of-fold confusion matrix for Model-1A DenseNet-121.

1) *Matched Architecture Comparison:* DenseNet-121 and ResNet-50 used the same cases, folds, augmentation, initialization policy, optimizer, epoch budget, and model-selection procedure. ResNet-50 achieved mean accuracy 0.8253 and

macro F1 0.8211. The paired DenseNet-minus-ResNet accuracy difference was 0.1064 (95% paired-bootstrap CI 0.0981–0.1153); the macro-F1 difference was 0.1098 (0.1013–0.1182).

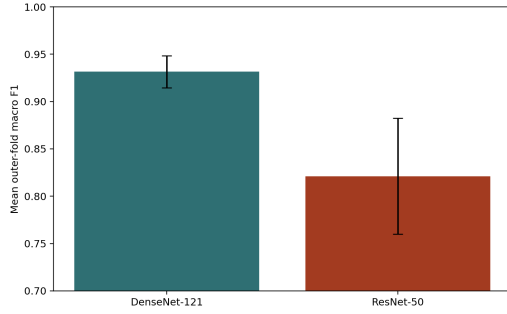


Figure 7. Controlled Model-1A comparison on matched grouped folds.

C. Model-1B Chest X-ray Results

DenseNet-121 completed all five patient-wise outer folds. The out-of-fold file contains all 112,120 images from 30,805 patients exactly once, with zero patient overlap. Mean macro AUROC was 0.7746 with standard deviation 0.0044 and a fold-bootstrap 95% interval of 0.7716–0.7780. Mean micro AUROC was 0.7932. At threshold 0.5, mean macro F1 was 0.1832; inner-selected thresholds increased mean macro F1 to 0.2229 and mean micro F1 to 0.2847.

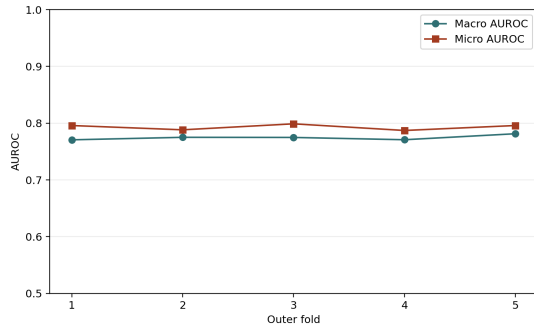


Figure 8. Model-1B macro and micro AUROC across patient-wise folds.

Pooled per-label AUROC ranged from 0.6566 for Nodule to 0.8819 for Edema. Nodule, Infiltration, Pneumonia, Mass, and Pleural Thickening were the weakest labels. These results are affected by prevalence, co-occurrence, and the automatically mined ChestXray14 labels.

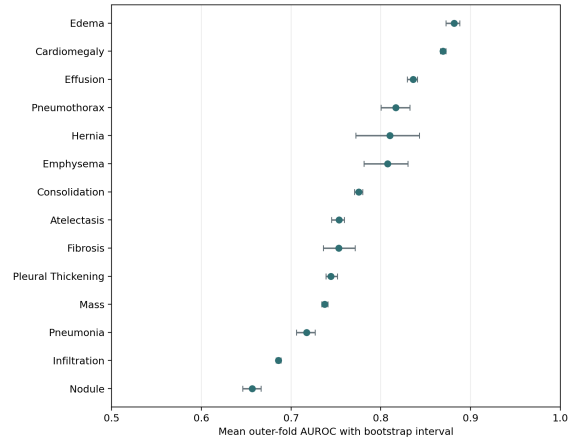


Figure 9. Model-1B per-label mean AUROC with fold-bootstrap intervals.

Pairwise Phi coefficients were calculated from all 112,120 metadata rows to describe target co-occurrence [23]. The strongest positive coefficients were Pneumothorax–Emphysema (0.1782), Pneumonia–Edema (0.1740), and Atelectasis–Effusion (0.1724). They describe annotation association, not classifier performance or clinical causality.

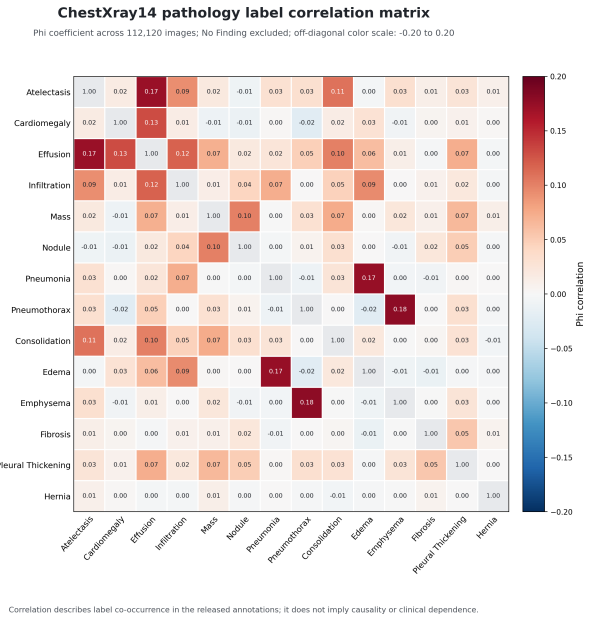


Figure 10. Pairwise Phi coefficients for the fourteen ChestXray14 pathology labels.

Table XIII
MODEL-1B PATIENT-WISE FIVE-FOLD RESULTS.

Fold	Images	Patients	Macro AUROC	Micro AUROC	Macro F1, tuned	Micro F1, tuned
1	22,639	6,161	0.7707	0.7959	0.2223	0.2829
2	22,103	6,161	0.7751	0.7883	0.2267	0.2763
3	22,221	6,161	0.7749	0.7989	0.2230	0.2909
4	22,770	6,161	0.7708	0.7872	0.2183	0.2904
5	22,787	6,161	0.7814	0.7959	0.2241	0.2829
Mean $\pm$ SD	112,120 OOF	30,805 unique	0.7746 $\pm$ 0.0044	0.7932 $\pm$ 0.0052	0.2229 $\pm$ 0.0031	0.2847 $\pm$ 0.0061
Fold-bootstrap 95% CI	–	–	[0.7716, 0.7780]	[0.7891, 0.7971]	[0.2203, 0.2252]	[0.2803, 0.2892]

Table XIV
PROTOCOL-AWARE CONTEXT FOR PUBLISHED CHEST-RADIOGRAPH RESULTS.

Study	Protocol	Reported result	Reason not directly comparable
This thesis	Five patient-wise outer folds with patient-wise inner validation; fourteen labels.	Macro AUROC 0.7746.	Internal reference; random initialization and no external institution.
Wang et al. [9]	Random image 70/10/20 split; earlier eight-label release.	Mean of reported AUROCs 0.7212.	Different release, label set, and split unit.
Baltruschat et al. [10]	Five patient-confined repeated holdouts; ImageNet experiments.	Mean pathology AUROC 0.821 ± 0.012 .	Repeated holdout, preprocessing, loss, and initialization differ.
Kufel et al. [11]	Custom patient-wise split; EfficientNet-B1 with ImageNet initialization.	Mean AUROC 0.843.	Different split, backbone, and initialization.

D. Model-1 Published Benchmark Context

Tables XII and XIV compare protocols as well as reported values. The reviewed studies differ in dataset version, split construction, duplicate control, initialization, augmentation, thresholds, and metric definitions. Their numbers are contextual; no significance test or superiority claim is made against published summaries.

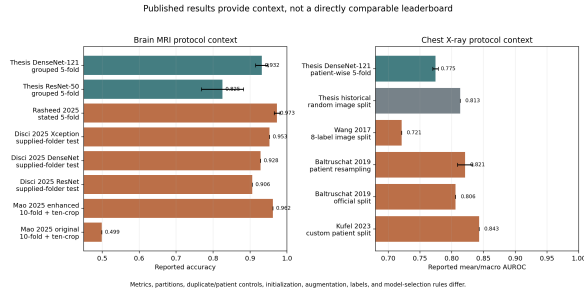


Figure 11. Contextual literature values with source-specific protocols and metric definitions.

E. Model-1 Error and Calibration Analysis

Mean Brain MRI Brier score was 0.1030 and mean ten-bin expected calibration error was 0.0262. These values measure probability calibration within the grouped folds. For Chest X-ray, changing thresholds alters F1 but not AUROC. The error summary in Figure 12 records the main class- and label-level weaknesses.

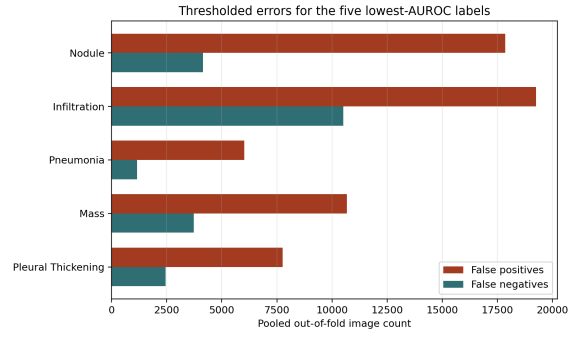


Figure 12. Error analysis from pooled Model-1 out-of-fold predictions.

F. Model-2 Results

1) *Model-2A Scanned Documents*: The pipeline processed 241 records. OCR completion was 1.0000, entity-field completion was 0.8755, and structured-output completion was 1.0000. These values show that the implemented stages produced their expected artifacts. They do not measure transcription correctness, field-level precision or recall, or clinical interpretation.

2) *Model-2B Doctor-Typed Text*: The specialty classifier was evaluated on 969 held-out records from the 29-specialty split. Test accuracy was 0.3209, macro F1 was 0.3468, and weighted F1 was 0.2962. The macro F1 exceeded accuracy by 0.0259, showing that the result was not defined only by the largest specialty, although overall discrimination remained limited. The output is a specialty prediction and should not be described as general clinical-text understanding.

The entity experiment produced token accuracy 0.9937 and entity F1 0.0148. Most tokens are not entities, so token accuracy is dominated by true negatives. The entity-level result is the more informative value and shows that the weak-label extraction component is not reliable enough to support an expert entity-recognition claim.

A recent leakage-free MTSamples preprint reported macro F1 of 0.279 for CLiGNet across 40 specialties and 4,966 records [29]. The present 0.3468 result is numerically higher, but it uses 29 retained specialties and a different split; therefore, it is a contextual reference rather than evidence of superiority. Earlier peer-reviewed clinical-subdomain studies also used different class sets, institutions, annotations, and metrics, which prevents a direct numerical ranking [15]. Model-2A is not placed in an OCR accuracy leaderboard because its 241 records lack expert-corrected transcripts and field labels.

G. Model-3 Paired IU Results

1) *Locked-Test Headline Results*: The locked set contains 352 studies. Fusion had the highest point estimate for macro AUROC, macro AUPRC, macro F1, micro F1, and label-wise accuracy. The image branch supplied most of the predictive signal; the Indication-only text branch was substantially weaker.

Table XV
LOCKED-TEST RESULTS FOR THE PAIRED IU EXPERIMENT ($n = 352$ STUDIES).

Model	Macro AUROC	Macro AUPRC	Macro F1	Micro F1	Bal. acc.	Label acc.
Image only	0.7910	0.3064	0.2915	0.3173	0.6948	0.8264
Text only	0.6282	0.1257	0.1442	0.1887	0.5532	0.7753
Fusion ($\alpha = 0.75$)	0.7954	0.3107	0.3071	0.3295	0.6875	0.8509

Fusion micro AUROC was 0.8280, micro AUPRC 0.3025, macro precision 0.2272, macro recall 0.5052, and exact-match accuracy 0.3381. The all-negative baseline achieved label-wise accuracy 0.9347 because 93.47% of label slots were negative, but it had zero positive recall. Label-wise accuracy is therefore not used as the principal measure.

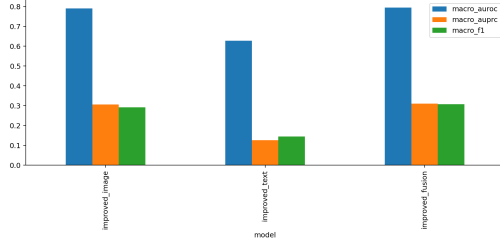


Figure 13. Locked-test comparison of image-only, Indication-only text, and late fusion.

2) *Paired Uncertainty*: Fusion-minus-image differences were $+0.0044$ macro AUROC (95% CI -0.0042 to 0.0133), $+0.0043$ macro AUPRC (-0.0186 to 0.0227), and $+0.0156$ macro F1 (-0.0128 to 0.0444). All three intervals crossed zero. Fusion therefore has modestly higher point estimates, but the locked cohort does not establish a statistically clear improvement over image-only.

Fusion-minus-text differences were larger: $+0.1672$ macro AUROC (0.1245 – 0.2098), $+0.1850$ macro AUPRC (0.1384 – 0.2452), and $+0.1630$ macro F1 (0.1148 – 0.2104).

Table XVI
LOCKED-TEST PER-LABEL RESULTS FOR THE SELECTED FUSION MODEL.

Label	Support	AUROC	AUPRC	F1	Precision	Recall
Cardiomegaly	43	0.9121	0.5642	0.5426	0.4070	0.8140
Pulmonary Atelectasis	40	0.7776	0.2727	0.3793	0.2895	0.5500
Calcified Granuloma	34	0.7618	0.2599	0.2897	0.1892	0.6176
Cicatrix	24	0.8290	0.5393	0.4063	0.3250	0.5417
Pleural Effusion	17	0.9240	0.4930	0.4314	0.3235	0.6471
Atherosclerosis	17	0.8460	0.2344	0.2400	0.1818	0.3529
Airspace Disease	16	0.7917	0.2166	0.2222	0.1724	0.3125
Scoliosis	12	0.7787	0.2801	0.2308	0.1500	0.5000
Granulomatous Disease	15	0.6633	0.0861	0.1414	0.0833	0.4667
Nodule	12	0.6696	0.1609	0.1875	0.1500	0.2500

3) *Per-Label Fusion Performance*: Cardiomegaly and Pleural Effusion had the strongest fusion discrimination. Nodule and Granulomatous Disease remained weak, with low support and low F1. Aggregate metrics should not conceal these failures.

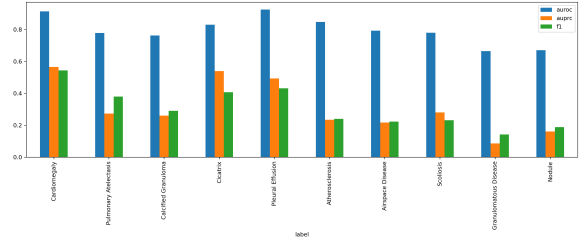


Figure 14. Per-label locked-test performance for the image, text, and fusion branches.

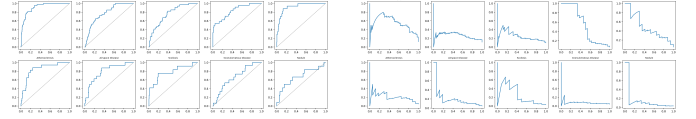


Figure 15. Per-label fusion ROC curves (left) and precision-recall curves (right).

4) *Development Selection and Diagnostics*: The fusion weight $\alpha = 0.75$ was chosen using development predictions. Figure 16 shows that selection; it is not a test-set sweep. The final image seeds each ran the fixed 16-epoch schedule. Development fold histories varied from 13 to 18 epochs because of early stopping, and only two of fifteen fold runs attained their best macro AUPRC at the stopping epoch.

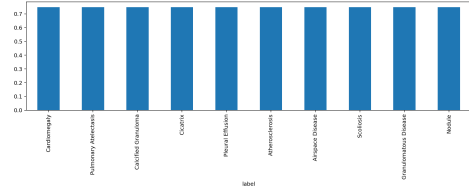


Figure 16. Development-only selection of the global image weight.

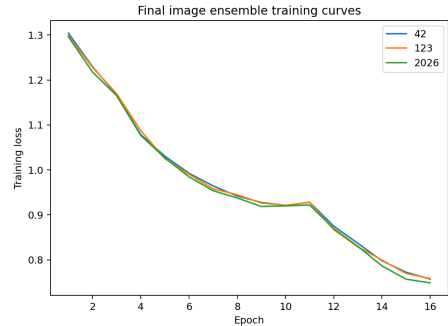


Figure 17. Selected IU image-branch development training histories.

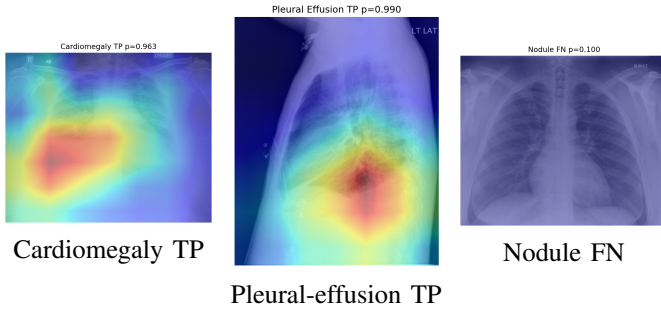


Figure 18. Frozen Grad-CAM diagnostic cases. Heatmaps show model sensitivity, not lesion segmentation or causal explanation.

5) Findings-Generation Results and Published Context:

The report-generation component achieved ROUGE-1, ROUGE-2, and ROUGE-L scores of 0.4569, 0.3692, and 0.4895, respectively. These values were calculated by comparing generated Findings with the reference Findings from the 352-study locked test set. ROUGE was reported separately because it measures lexical text overlap rather than diagnostic classification performance or clinical correctness.

ROUGE-L is commonly reported in IU radiology report-generation research. R2Gen reported 0.371, the cross-modal memory network reported 0.375, and SERPENT-VLM reported 0.452 [19]–[21]. The value obtained here is numerically higher, as shown in Table XVII, but it is not a direct superiority result. The published systems generate reports under their own IU partitions and preprocessing, whereas this thesis generates the Findings section from a fixed ten-label template process on a 352-study locked set. Differences in target text, cohort filtering, tokenization, and ROUGE implementation can materially change the score.

Table XVII

PROTOCOL-QUALIFIED IU REPORT-GENERATION CONTEXT. ONLY ROUGE-L IS SHOWN BECAUSE THE CITED PAPERS REPORT BLEU-1/2 RATHER THAN ROUGE-1/2.

System	ROUGE-L	Evaluation context
R2Gen [19]	0.371	Published IU report-generation protocol using a memory-driven transformer.
Cross-modal memory network [20]	0.375	Published IU report-generation protocol with cross-modal memory.
SERPENT-VLM [21]	0.452	Published IU report-generation protocol; reported test size 395.
Model-3 in this thesis	0.4895	Template-based generated Findings; locked test of 352 eligible same-study records.

Chapter Summary

Model-1 produced strong leakage-aware internal results under grouped and patient-wise evaluation. Model-2 produced a held-out specialty result and bounded processing evidence, but its document and weak-label entity outputs cannot support expert semantic claims. Model-3 produced the highest fused classification point estimates and a locked-test ROUGE-L score of 0.4895; protocol differences prevent direct superiority

claims against published report generators. No result in this chapter establishes external or clinical validity.

VIII. DISCUSSION, LIMITATIONS, CONCLUSION, AND FUTURE WORK

A. Discussion

The results support three separate technical systems rather than one fused end-to-end claim. Model-1 provides strong leakage-aware internal image results, Model-2 provides bounded document-processing and typed-text evidence, and Model-3 provides a same-study IU evaluation of its own image and Indication branches, late fusion, and Findings generation. The evidence is retrospective and task-specific, so none of these results should be interpreted as clinical validation.

B. Integrated Interpretation

Model-1 demonstrates how the partition unit affects the credibility of medical-image performance. Brain MRI DenseNet-121 achieved mean grouped-fold accuracy of 0.9317 and macro F1 of 0.9316. On the same cases and training policy, it outperformed the matched ResNet-50 comparison. Patient identifiers were unavailable, however, so similarity grouping is a conservative leakage control rather than proof of patient independence. Chest X-ray DenseNet-121 achieved mean patient-wise macro AUROC of 0.7746 and tuned macro F1 of 0.2229. The weaker results for Nodule, Infiltration, and other labels show why one average cannot represent all disease categories.

Model-2A completed OCR and structured export for the reported document set, but completion does not measure transcription or field correctness. Model-2B achieved accuracy of 0.3209 and macro F1 of 0.3468 for 29-specialty classification. Its entity F1 of 0.0148 against weak labels is insufficient for a reliable clinical entity-extraction claim. These findings justify retaining Model-2 as an independently evaluated technical system with explicit annotation limits.

Model-3 closes the pairing problem that affected the earlier architecture. Its radiographs and Indication text come from the same IU study, while its image and text submodels are trained specifically for the paired task. Fusion achieved macro AUROC of 0.7954, macro AUPRC of 0.3107, and macro F1 of 0.3071. These point estimates exceed the image-only values of 0.7910, 0.3064, and 0.2915, but the paired fusion-minus-image intervals cross zero. The correct conclusion is therefore a modest, uncertain incremental contribution from Indication text rather than proven fusion superiority.

The template-generation component achieved ROUGE-1 of 0.4569, ROUGE-2 of 0.3692, and ROUGE-L of 0.4895 on the locked test set. Its ROUGE-L value is numerically above the published IU values included in Chapter 7, but differences in cohort construction, output section, tokenization, and generation method prevent a direct ranking. The score validates reproducible lexical comparison with hidden Findings; it does not establish clinical completeness or factual correctness.

C. Limitations

1) *Clinical and External Validity*: No system has undergone prospective evaluation, independent-hospital testing, or structured review by qualified clinicians. The study does not measure diagnostic safety, treatment effect, workflow benefit, or patient outcomes. The software is not approved or intended for patient care.

2) *Brain MRI Data*: Patient identifiers are unavailable. Conservative perceptual hashing keeps detected similarity groups together, but it cannot determine whether every candidate is the same acquisition or whether undetected related images remain. All evaluation is internal to one local representation of the dataset.

3) *ChestXray14 Data*: The corrected folds are patient-wise, but the labels are automatically mined and affected by imbalance, co-occurrence, and possible omission. Both AP and PA views are included. Thresholded results depend on inner-validation operating points, and the corrected protocol was not repeated on an external cohort.

4) *Model-2 References*: Model-2A lacks expert-corrected OCR transcripts and field-level annotations for the reported 241 records. Completion cannot detect an incorrect medicine, laboratory value, unit, or date. Model-2B specialty results apply only to the retained MTSamples-derived classes. Its entity labels come from rules and dictionaries rather than clinical adjudication.

5) *IU Pairing and Labels*: The IU cohort is small, retrospective, and internal. A reliable patient identifier is unavailable in the accessed representation, so separation is study-wise and repeat patients cannot be excluded. Problems labels can be incomplete or differently scoped relative to Findings, and rare targets make AUPRC and threshold-dependent estimates unstable. Indication provides limited clinical context.

6) *Statistical and Literature Comparison*: Bootstrap intervals measure sampling uncertainty within the available predictions; they do not capture dataset shift. Published benchmarks use different data versions, partitions, preprocessing, initialization, output targets, and metric implementations. Their values provide context rather than a statistically controlled comparison. ROUGE measures lexical overlap and may reward common template wording even when clinically important details are absent.

D. Conclusion

This thesis developed and evaluated three separate systems. Model-1 is a standalone medical-image classifier supported by complete grouped Brain MRI and patient-wise Chest X-ray out-of-fold predictions. Model-2 is a standalone document and doctor-typed-text system whose specialty classifier has a held-out result, while its OCR and entity outputs require narrower interpretation. Model-3 is a new IU-specific architecture containing its own image ensemble, Indication-text classifier, development-selected late fusion, and template-based Findings generator.

Model-3 produced the highest fused classification point estimates on its locked 352-study test set and achieved ROUGE-

L of 0.4895 for generated Findings. The uncertainty analysis does not establish a clear fusion advantage over image-only, and the generation metric does not establish clinical correctness. The contribution is therefore a reproducible retrospective technical evaluation with explicit evidence boundaries, not a clinically validated diagnostic system.

E. Future Work

1) *Expert Annotation for Documents and Typed Text*: Future work should create expert-corrected OCR transcripts, field-level document annotations, and clinician-reviewed text entities. Evaluation should include character and word error rates, field precision and recall, negation, temporality, and inter-annotator agreement.

2) *External Image Evaluation*: Both Model-1 tasks should be tested at independent institutions with predefined inclusion criteria, patient-level separation, duplicate control, subgroup analysis, calibration, and confidence intervals. A fair Chest X-ray architecture comparison should repeat the complete patient-wise protocol.

3) *Larger Paired Cohorts*: Model-3 should be repeated on datasets with stable patient identifiers and institutionally independent test sites. Larger positive supports would permit more reliable per-label estimates. Fusion choices, templates, and endpoints should be frozen before external test labels are opened.

4) *Clinically Reviewed Findings Generation*: Future generation experiments should compare template and neural approaches under the same cohort and evaluation code. In addition to ROUGE, evaluation should include clinical entity agreement, negation errors, omissions, unsupported findings, and blinded review by radiologists. Any added context must be available at inference time and must not expose the reference Findings or Impression.

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